

Pharmacognostic and phytochemical studies on *Ficus Microcarpa* L. fil

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ABSTRACT

Background: *Ficus microcarpa* L. fil. (*Syn: Ficus retusa*) (Moraceae) is well-known traditional medicinal plant. The bark is used for diverse health ailments in traditional and folklore remedies.

Aims: The present study was undertaken to lay down pharmacognostical and phytochemical standards.

Materials and Methods: Pharmacognostic studies on fresh, dried and powdered bark was carried out to determine its morphological, anatomical, and phytochemical diagnostic features. Furthermore, major phytoconstituents were identified from the extracts with the help of high performance liquid chromatography (HPLC) study.

Results: The morphology showed to be soft, brittle, rough, shallow vertical, irregularly oriented fissures, curved surface; with splintering, laminated fracture. Microscopically *F. microcarpa* showed all general characteristics of bark with some distinct differentiation. The phellem is thin and even, phelloderm few cell and rectangular and followed by smaller sclerides, the phloem rays are broad, multi-serrate and showed the scattered bundles of sclerides. The fluorescence and physicochemical standards for bark were established. HPLC analysis showed the predominant presence of therapeutically important phytoconstituents such as oleanolic, betulinic acid, lupeol, β -sitosterol, catechin, and gallic acid.

Conclusion: The bark of *F. microcarpa* considered equivalent to other *Ficus* species, such as *Ficus virens*, *Ficus infectoria*, *Ficus arnottiana*, *Ficus lacor*, and *Ficus talboti*. However, there is no pharmacognostical and phytochemical reports on *F. microcarpa* to authenticate and differentiate it from similar species. Present work has described pharmacognostical and phytochemical characteristics of *F. microcarpa* and diagnostic features to differentiate it.

KEY WORDS: *Ficus microcarpa*, *Ficus retusa*, pharmacognostic, phytochemical study

and bark are used in diverse ailments in traditional and folklore remedies. The phytochemistry of the plant is well-studied and the literature supports the presence of triterpenoids, fatty alcohol, steroids, coumarin, flavane, 4-hydroxybenzoates, and a carotenoid-like compound in *Ficus microcarpa* bark (FMB).^[1] The antioxidant and antibacterial potential plant phenolics such as protocatechuic acid, catechol, *p*-vinylguaicol, syringol, *p*-propylphenol, vanillin, and syringaldehyde have been identified in FMB.^[2] The latex of the plant has been reported to contain chitinase, giving rise to antifungal properties.^[3] Liu *et al.*, reported antitussive and expectorant potential of *F. microcarpa*.^[4] Traditionally, the bark has a reputation of being efficient in the treatment of diabetes, ulcers, burning sensation, hemorrhages, leprosy, itching, liver disease, and toothache.^[5] Despite extensive traditional utilization and source of important phytoconstituents, the present plant has not been well standardized for pharmacognosy and phytochemistry. The present study was undertaken to lay down pharmacognostical and phytochemical standards for *F. microcarpa*.

MATERIALS AND METHODS

FMB was collected from Lonawala, Maharashtra, in August 2009 and authenticated by the Botany Department of SSVPS College, Dhule, Maharashtra. The bark was washed, cleaned, shade dried, powdered, and passed through a 40-mesh sieve. It was stored in a tightly closed container. A voucher specimen R C Patel (it is the college code for herbarium library) (RCP-42) was authenticated and deposited in Pharmacognosy Department.

INTRODUCTION

Ficus microcarpa L. fil. (*Syn: Ficus retusa*) (Moraceae) is well-known traditional medicinal plant known commonly known as Indian Laurel, distributed across the hotter parts of India. Its dried leaves, aerial roots,

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Macroscopy

The bark was separated from other parts, washed, cleaned, and dried for further use. The following macroscopic characters of the fresh leaves were noted: Color, odor, taste, size and shape, touch, texture, and fracture [Figure 1].

Microscopy

The free hand thin transverse sections of the fresh bark taken and treated with the different staining agent and observed for the general and specific microscopic characteristic [Figure 1]. Furthermore, small quantity of the powdered

leaves was cleared, mounted, and observed for diagnostic powder characteristics [Figure 2]. Photomicrographs of the microscopical sections were taken with the help of Motic photomicroscope provided with Motic Images plus 2.0 software.

Physicochemical and fluorescence analysis

Bark powder was analyzed for their fluorescence analysis according to Kalaskar *et al.*^[6] The dried bark material was analyzed under visible light, short, and long ultra violet (UV) light after treatment with various acids, alkalis, other reagents [Tables 1 and 2]. Physicochemical parameters such as a percentage of ash, extractive values, foreign matter content, and loss on drying were determined in triplicates as per Kalaskar and Surana.^[7] The mean and standard deviation ($\pm$ SD) from triplicate samples were summarized in Table 3.

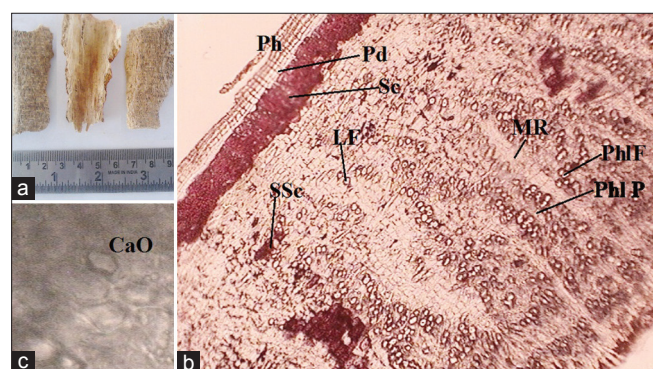


Figure 1: *Ficus microcarpa*: (a) External features of bark. (b) Transverse section of bark. (c) Enlarged portions. Ph = Phellem, Pd = Phelloderm, Sc = Sclereids, LF = Lignified fibers, SSc = Scattered Sclereids, MR = Medulary rays and Phl F-Phloem fiber; Phl P-Phloem Parenchyma

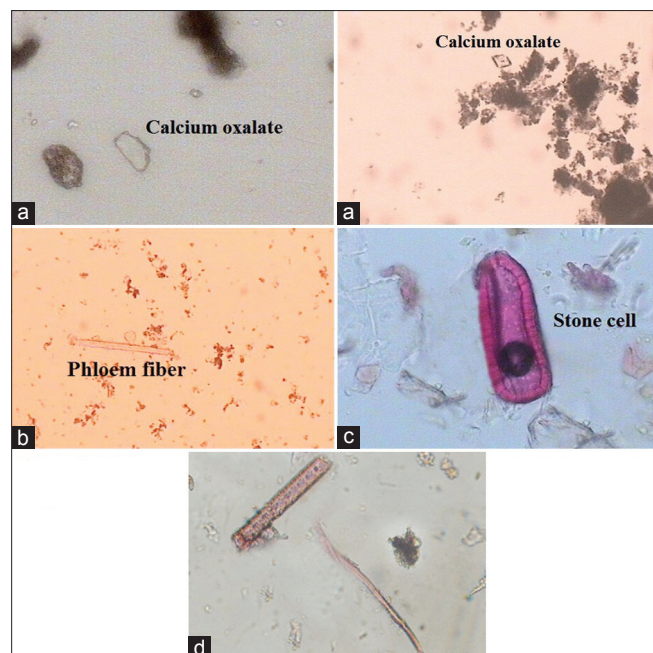


Figure 2: Powder microscopy of *Ficus microcarpa* bark. (a) Calcium oxalate crystal. (b) Phloem fiber. (c) Stone cell/sclereids. (d) Lignified fibers

Table 1: Fluorescence analysis of *Ficus microcarpa* bark

Color reaction	Day light	UV light 365 nm
Powder+NaOH in methanol	Yellowish brown	Yellowish fluorescence
Powder+NaOH in methanol+nitrocellulose in acetic acid	Yellow brown	Dark yellowish fluorescence
Powder+nitrocellulose in acetic acid	Grayish green	Strong yellow fluorescence
Powder+NaOH in water	Red brown	Faint green fluorescence
Powder+nitrocellulose acetic acid+HCl	Yellow fluorescence	Yellow fluorescence
Powder+HCl	Yellow	Dark yellow with faint yellow fluorescence
Powder+H ₂ SO ₄	Buff	Faint yellow
Powder+HNO ₃	Yellow	Yellowish purple
Powder	Buff yellow	Slight pink

Table 2: Behavior of *Ficus microcarpa* bark powder with different chemical reagents

Reagents	Color/precipitation	Constituents
Picric acid	No precipitations	Alkaloids absent
Conc. H ₂ SO ₄	Reddish brown	Steroids/triterpenoids present
Aq. FeCl ₃	Blackish brown	Tannins present
Iodine solution	Blackish	Starch present
5% Aq. KOH	No change	Anthroquinone glycosides absent
Mayer's reagent	No perception	Alkaloids absent
Spot test	Stains observed	Fixed oils present
Aq. AgNO ₃	No precipitation	Proteins absent
Aq. NaOH	Yellow	Flavonoids present
Mg-HCl	Magenta	Flavonoids present
Dragendroff's reagent	No ppt	Alkaloids absent

Phytochemical analysis

The FMB extracted with four successive solvent to get different polarity solvents, to screen phytoconstituents based on their polarity. These extracts were evaluated for qualitative phytochemical analysis. Furthermore, the presence of triterpenoids, steroids, and polyphenols in FMB extract were estimated qualitatively by HPLC, as per the method described by Pawar and Surana,^[8] with some modification. A Shimadzu HPLC system with LC-10AT, UV detector (Spectra System UV 1000), and Luna C₁₈ reverse-phase column (250 mm × 4.6 mm, i.d. particle size 5 μ) in an isocratic mode was used with the specification given in Table 4.

RESULTS

Macroscopic features

The bark has a thickness of about 4-7 mm and grayish to dark brown outside and light brown inside. It is soft, brittle,

Table 3: Physicochemical analysis of *Ficus microcarpa* bark

Types of ash value/extractive values	% w/w
Ash values	
Total ash	17.9±1.01
Acid insoluble ash	7.69±0.62
Water soluble ash	5.14±0.46
Sulphated ash	3.99±0.27
Extractive values	
Petroleum ether 60-80°	2.71±0.29
Ethyl acetate	1.37±0.26
Alcohol	7.08±0.27
Water	3.35±0.26
Foreign organic matter	1.59±0.33
Loss on drying	4.55±1.59

Values are mean±SD, n=3

Table 4: HPLC specification for phytochemical analysis of FMB extract

Phytoconstituents	Mobile phase	Flow rate (ml)	Rt (min)		Detection (nm)
			Test	Standard	
Oleanolic acid	Acetonitrile: Water (85:15, v/v)	0.3	PFM-21.37 EAFM-22.20	22	215
Betulinic acid			PFM-18.09 EAFM-19.42	18	215
Lupeol	Acetonitrile: Water (90:10, v/v)	1	PFM-6.10	6.9	350
β-sitosterol		0.3	PFM-11.16	11.06	210
Quercetin	0.5% formic acid: Acetonitrile (70:30)	0.9	EAFM-NA EOFM-NA	16.9	340
Catechin		0.9	EAFM-2.55 EOFM-2.56	3.6	280
Gallic acid	0.1% H ₃ PO ₄ : Acetonitrile (85:15)	0.7	EOFM-3.62 AQFM-3.63	3.8	280

PFM: Pet ether (60-80°) *Ficus microcarpa* extract, EAFM: Ethyl acetate *Ficus microcarpa* extract, EOFM: Ethanol *Ficus microcarpa* extract, AQFM: Aqueous *Ficus microcarpa* extract, HPLC: High performance liquid chromatography, FMB: *Ficus microcarpa* bark

with a rough surface with shallow vertical, irregularly oriented fissures, and curved. The fracture was found to be splintery and laminated [Figure 1].

Microscopic characteristics

Bark differentiated into outer bark as periderm and inner bark or secondary phloem. Periderm well differentiated into phellem (Ph) and phelloderm (Pd). Ph zone is thin and even in transaction. Ph cells are organized into thin tangential compressed cells, rectangular, and the older layers exfoliate. The periderm consists of continuous bands of sequential periderm, and originates from the secondary phloem. Ph cells are homogeneous thin walled rectangular and suberized while Pd is wide and distinct consist of 5-7 layers if containing chloroplast or tannin. Pd cells (23.87 μ × 13.45 μ) are turned into cubical sclereids (Sc) (49.51 μ × 21.24 μ) arranged 2-5 layers, lignified [Figure 1].

Secondary phloem is differentiated into inner intact non-collapsed zone and outer collapsed phloem zone. The collapsed phloem zone consists of wide dilated modular rays, few phloem fibers (17.70 μ), lignified fibers (25.75 μ), and cubical Sc are scattered in this region [Figure 1] In the non-collapsed zone phloem elements occur in small clusters and consist of phloem fibers, sieve tube members, companion cells, and phloem parenchyma. The phloem rays are uniseriate or multi-seriate. They are homocellular or heterocellular. Laticifers are found in the inner bark. Starch and prismatic calcium oxalate is abundant in most of the axial parenchyma cells.

Powder drug analysis

The microscopical powder analysis showed the presence of lignified phloem fibers, sclerides in the form of stone cells,

rhomboidal prisms of calcium oxalate and abundant starch grain in parenchymatous cells observed [Figure 2].

Physicochemical analysis

The fluorescence analysis and behaviors of powder with different chemical reagents were studied [Tables 1 and 2]. The physicochemical standards are important to check the quality, purity, and adulteration of given crude drug. The foreign matter, loss on drying (LOD), ash, and extractive values were determined and summarized in Table 3.

Phytochemical analysis

The preliminary phytochemical analysis of bark extracts showed the medicinally potential constituents include carbohydrates, tannins, flavonoids, steroids, and triterpenoids. The HPLC study of successive FMB extracts identified predominant phytoconstituents such as triterpenoids and steroids such as oleanolic acid (OA), betulinic acid (BA), lupeol, and β -sitosterol in pet-ether extract. Triterpenoids such as OA, BA and polyphenol, catechin (Cat) were identified in ethyl acetate extract. The ethanol extract showed the presence of polyphenols, Cat, and gallic acid (GA) while phenolic acid, GA was identified in aqueous extract [Table 4].

DISCUSSION

The bark is one of the important medicinal plant parts, used traditionally for therapeutic purposes. The bark is very complex in structure and has the potential of containing many primary and secondary metabolites. The complex structure of the bark can be utilized for correct identification to maintain the quality and purity of the drug.^[9] The different *Ficus* species are utilized extensively in Indian traditional medicinal systems. Among these species *F. microcarpa* having credentials in treatment of diabetes, ulcers, burning sensations, hemorrhages, leprosy, itching, liver disease, and toothache. The FMBs considered equivalent with other species such as *Ficus virens*, *Ficus infectoria*, *Ficus arnottiana*, *Ficus lacor* and *Ficus talboti*.^[10-12] Hence, it is always a problem to identify the correct species without sets of standards.

Thus, the present work was taken up with an objective to lay down detail pharmacognostical and phytochemical standards that will contribute significantly to quality control of medicinally useful *F. microcarpa*.

Microscopy of *F. microcarpa* showed all general characteristics

of bark with some distinct differentiation. The bark has yellowish brown to reddish brown color with very few distinct irregular shallow fissures. Microscopically the Ph is thin and even, which is not so in other species of *Ficus*.^[13] Furthermore, the Pd few cell and rectangular and followed by sclerides as stone cell layer. The sclerids are smaller when compared with other species.^[13] In the upper phloem region, the phloem rays are broad, multiserrate and showed the scattered bundles of sclerides. The phloem fibers are distinct and solitary, lignified. The phloem rays and parenchyma showed the presence of starch and rhomboidal calcium oxalate crystals.

The ash values of a drug give an idea of the earthy matter or inorganic composition and other impurities present along with the drug. Extractive values are useful for determination of exhausted or adulterated drug.^[6] Thus ash, extractive values, fluorescence analysis will be helpful in the identification and authentication of plant material. The preliminary phytochemical screening showed the presence of different phytoconstituents groups such as steroids, triterpenoids, flavonoids, polyphenolic, and tannin. HPLC analysis showed the predominant presence of therapeutically important phytoconstituents such as oleanolic, BA, lupeol, β -sitosterol, Cat, and GA.

The present study provides in-depth microscopical features, and identification of biologically active phytoconstituents, which also provide pharmacopoeia standards for easy identification of the *F. microcarpa* and hence differentiating it from closely related species.

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